

Campus Signal Bell

A campus notice desk that turns one staff update into clear information for affected riders.

RESEARCH VERDICT: CAUTION

The configured feature set is buildable inside the demo budget, but access to reliable campus service data remains unverified.

DOCUMENT MAP

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Executive summary

This recalculated demo plan keeps the confirmed notice workflow small. It tests staff publishing and rider comprehension before adding live tracking or prediction.

PROTOTYPE

\$450 to \$1,200

TIMELINE

5 to 7 weeks

DIFFICULTY

medium

RESEARCH VERDICT

caution

Problem definition

Campus transit confusion

- Students miss buses when temporary route changes are buried in several different campus pages.
- Transit staff repeat the same delay updates across signs, social posts, and support messages.

Target users

Students who rely on campus transit

Confirmed interview inputs

Who should get the first working version?

Students who ride every day

Can the prototype assume access to a live vehicle feed?

Yes

Which early result would convince you the idea works?

Less repetitive work for staff

Research summary

Broad foothold

Campus riders already have general trip planners, but temporary local changes still create a narrow communication problem. A useful prototype should improve the notice loop before attempting live

vehicle prediction.

General transit apps cover the broad trip-planning job [1]

Existing products already provide routes, arrivals, and service information for many agencies.

What it changes: The demo concept needs a campus-specific workflow advantage rather than another generic map.

A standard feed can separate schedule data from live updates [2]

A prototype can begin with scheduled route data and add real-time vehicle or alert data later.

What it changes: This supports a small notice-first prototype that does not require live tracking on day one.

The main demo risk is fragmented communication [3]

The supplied problem notes place the failure between staff publishing a change and riders noticing it.

What it changes: A shared publishing and alert workflow may matter more than adding another rider-only interface.

Focused configured-project pass

CAUTION

The configured feature set is buildable inside the demo budget, but access to reliable campus service data remains unverified.

Campus Signal Bell still has a plausible campus-specific opening, but the value rests on a simpler staff-to-rider notice loop rather than generic trip planning.

Generic service alerts already exist [4]

The illustrative competitor supports general transit notices.

What it changes: The chosen project must win on the campus publishing workflow, not on having alerts at all.

The notice-first prototype stays small [5]

A web prototype can use scheduled data and manually entered service changes.

What it changes: Live prediction can remain outside the first build.

Known research gaps

- Market: The visual demo does not perform live market research or validate willingness to pay.
- Regulatory constraints: The token-free walkthrough does not research a real campus or jurisdiction.

Selected product concept

A campus notice desk that turns one staff update into clear information for affected riders.

This canned project demonstrates the structure, comparison, and feature-selection flow without asking a model to generate anything.

Distinct approach

A staff-to-rider notice pipeline built around temporary campus changes.

Build constraints

Target platform

Desktop-first responsive web app

Deadline

Eight weeks

Team size

One student builder

Team skills

TypeScript and basic product design

Regulatory constraints

No sensitive student records

Accessibility needs

Keyboard navigation and readable service alerts

Existing systems

A sample GTFS feed and manual service notices

Revenue model

Internal campus tool

Other constraints

The demo must work without installing a mobile app

Chosen features

Unified service notice · core

The smallest usable version of Campus Signal Bell needs this capability.

Targeted rider alerts · recommended

The Sage recommends this once the core loop is working.

Functional requirements

FR-1 Unified service notice

The smallest usable version of Campus Signal Bell needs this capability.

- A test user can complete the Unified service notice workflow.

FR-2 Targeted rider alerts

The Sage recommends this once the core loop is working.

- A test user can complete the Targeted rider alerts workflow.

Nonfunctional requirements

Accessibility

Current service notices must remain readable without color or animation.

Measure: Keyboard-only review passes and every notice has text status.

Freshness

Published changes must appear promptly.

Measure: A saved test notice appears within 30 seconds.

Prototype plan and budget

ESTIMATED RANGE

\$450 to \$1,200

PROTOTYPE TIMELINE

5 to 7 weeks

Estimate assumptions

- One student builds the prototype with hosted web services.
- The first test uses sample transit data and manual notices.

Production plan and budget

ESTIMATED RANGE

\$4,000 to \$10,000

Estimate assumptions

- One campus deploys the service with a maintained data feed.

Suggested technology

Application: TypeScript web app with a small relational database
It matches the demo builder skills and keeps the publishing workflow inspectable.

Hardware and manufacturing

No dedicated hardware or manufacturing work is required for this configuration.

Competitor comparison

Competitor	Overlap	Differentiation
Illustrative general transit app direct [4]	Both present service alerts to riders.	The selected project centers the campus staff publishing loop.

Differentiation and positioning

A staff-to-rider notice pipeline built around temporary campus changes.

Illustrative general transit app [4]
The selected project centers the campus staff publishing loop.

Risks and assumptions

Campus data access may be delayed or unavailable. [5]
Mitigation: Prototype with scheduled data and manual notices before integrating a live feed.

Technical difficulty
medium: The interface is routine, but reliable transit data and timely publishing need early testing.

Validation plan

1. One publishing action reduces repeated staff updates.

Method: Run a tabletop route-change exercise with two staff members.

Success signal: Both publish the complete notice without copying it into another tool.

High-level development phases

01 Notice loop prototype

Prove the staff-to-rider workflow.

- Staff notice form
- Rider notice page
- Sample campus route data

02 Small campus test

Measure comprehension and staff effort.

- Usability notes
- Timing measurements
- Revised requirements

Sources and citations

Generated Sep 4, 2026. Research retrieval dates appear with each source. This token-free demo uses illustrative sources and is not live research.

[1] Illustrative transit product comparison

Token-free demo tape · date unknown · retrieved Sep 4, 2026 · broad pass
Canned example showing how a competitor source will appear.

<https://example.com/demo/transit-apps>

[2] Illustrative transit feed documentation

Token-free demo tape · date unknown · retrieved Sep 4, 2026 · broad pass
Canned example showing a technical source and its evidence summary.

<https://example.com/demo/transit-feed>

[3] Illustrative campus notice observations

Token-free demo tape · date unknown · retrieved Sep 4, 2026 · broad pass
Canned example derived from the supplied demo problem.

<https://example.com/demo/campus-notices>

[4] Illustrative configured competitor page

Token-free demo tape · date unknown · retrieved Sep 4, 2026 · focused pass
Canned example showing how direct feature overlap appears.

<https://example.com/demo/configured-competitor>

[5] Illustrative implementation and cost notes

Token-free demo tape · date unknown · retrieved Sep 4, 2026 · focused pass
Canned example showing how feasibility evidence appears.

<https://example.com/demo/configured-feasibility>